

CSMA 2026

Efficient multidisciplinary design via Bayesian Optimization (BO)

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POLYTECHNIQUE
MONTREAL

UNIVERSITÉ
D'INGÉNIERIE



ONERA

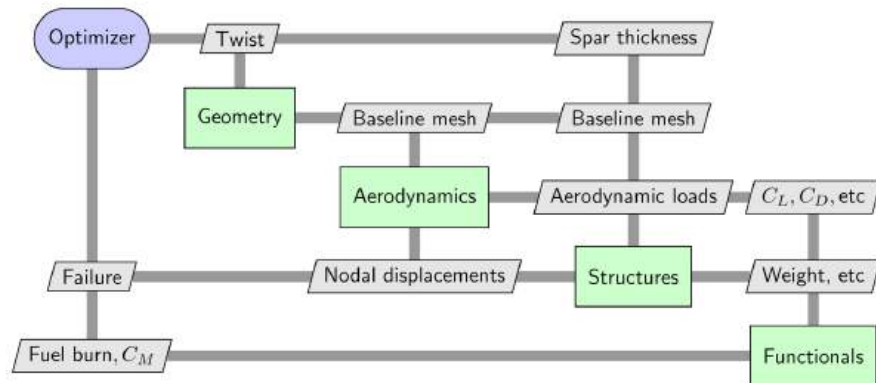
THE FRENCH AEROSPACE LAB

MDO (each green box is a costly simulation code)

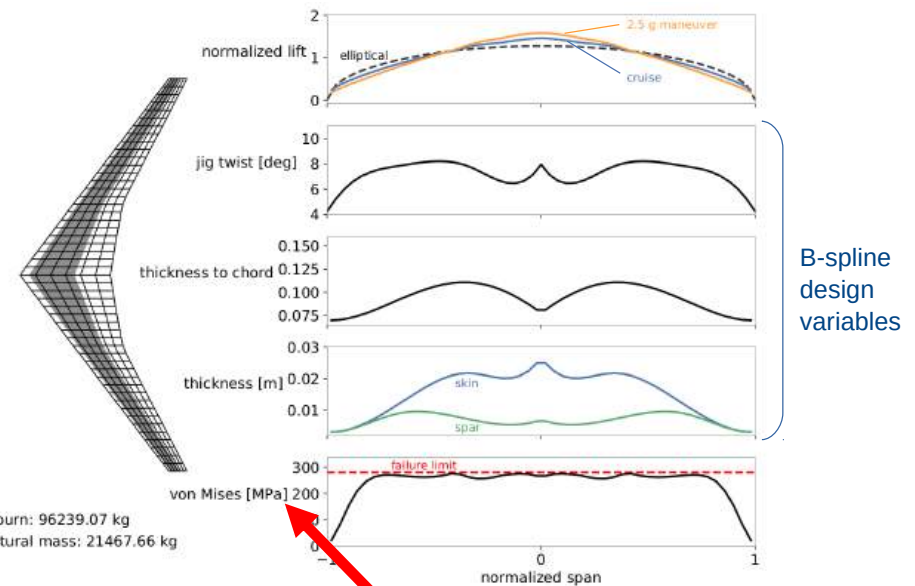
needs derivative (coupled), gradient based, scalable, need a good x^0

Results

This is MDO
Discipline Structure is coupled to
Discipline Aerodynamic !



Minimize fuel burn



fuel burn: 96239.07 kg
structural mass: 21467.66 kg

Wrt

$L=W$

Admissible stress

Etc...

Jasa, J. P., Hwang, J. T., & Martins, J. R. (2018). Open-source coupled aerostructural optimization using Python. *Structural and Multidisciplinary Optimization*, 57(4), 1815-1827.

Chauhan, S. S., & Martins, J. R. (2018, September). Low-fidelity aerostructural optimization of aircraft wings with a simplified wingbox model using OpenAeroStruct. In *International conference on engineering optimization* (pp. 418-431). Cham: Springer International

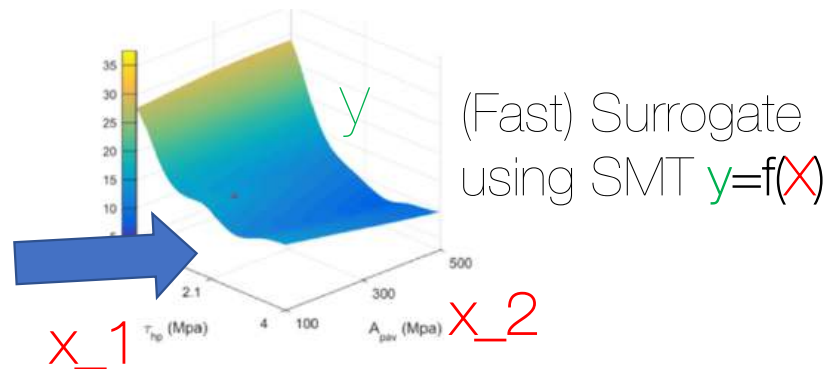
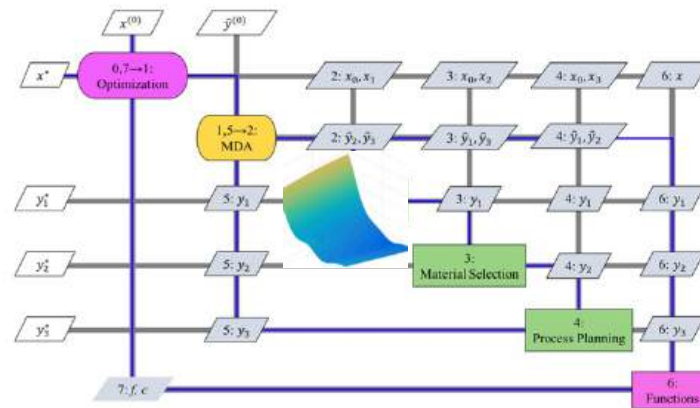
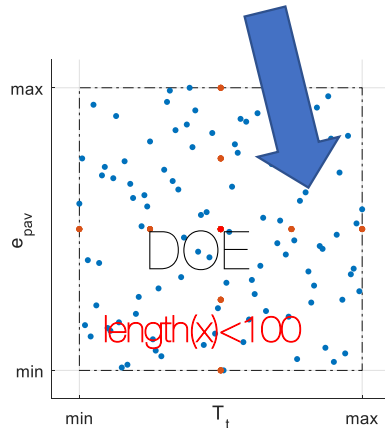
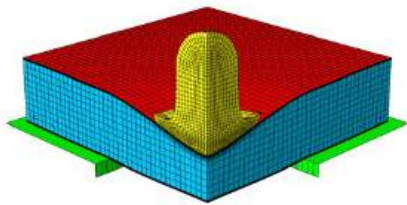
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Efficient MDO (means with Surrogate models)

A. Forrester, A. Sobester, and A. Keane. "Engineering Design via Surrogate Modelling: A Practical Guide". Coll. John Wiley & Sons (2008)

Dubreuil, S., Bartoli, N., Gogu, C., & Lefebvre, T. (2020). Towards an efficient global multidisciplinary design optimization algorithm. Structural and Multidisciplinary Optimization, 62(4), 1739-1765..

(Expensive) simulation



- A surrogate model of a function is an **approximation** of Expensive Computer simulation: It's a supervised learning process in AI.
- The surrogate can be used as a "super fast" code in a Multidisciplinary Design Optimization loop.
- Or do **Bayesian Optimization**

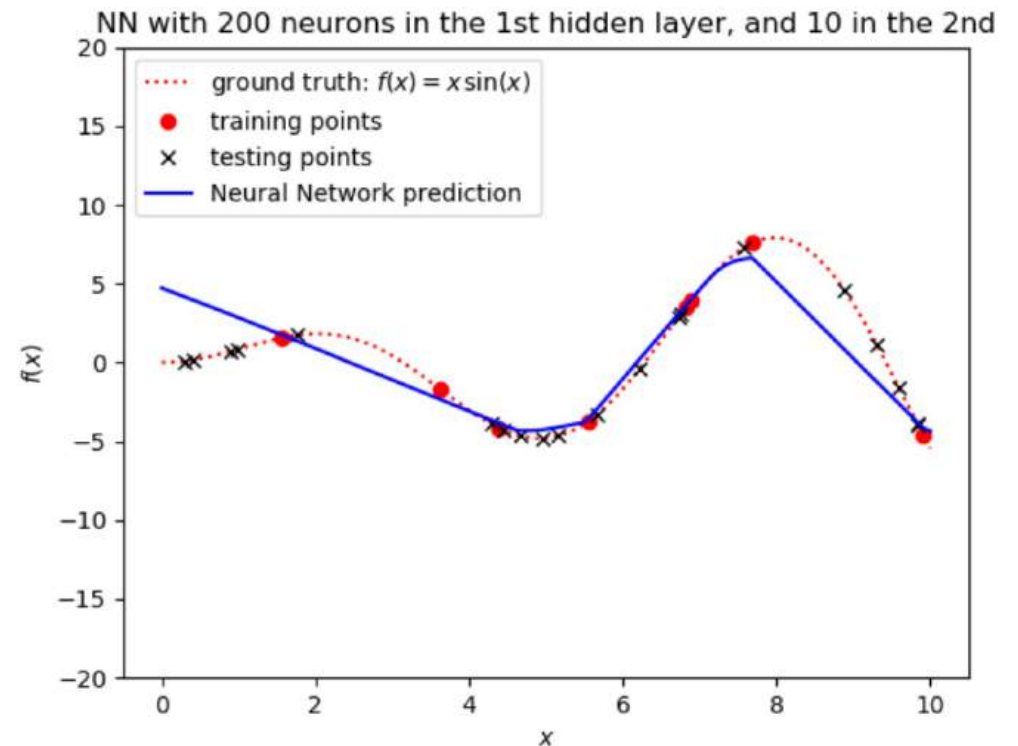
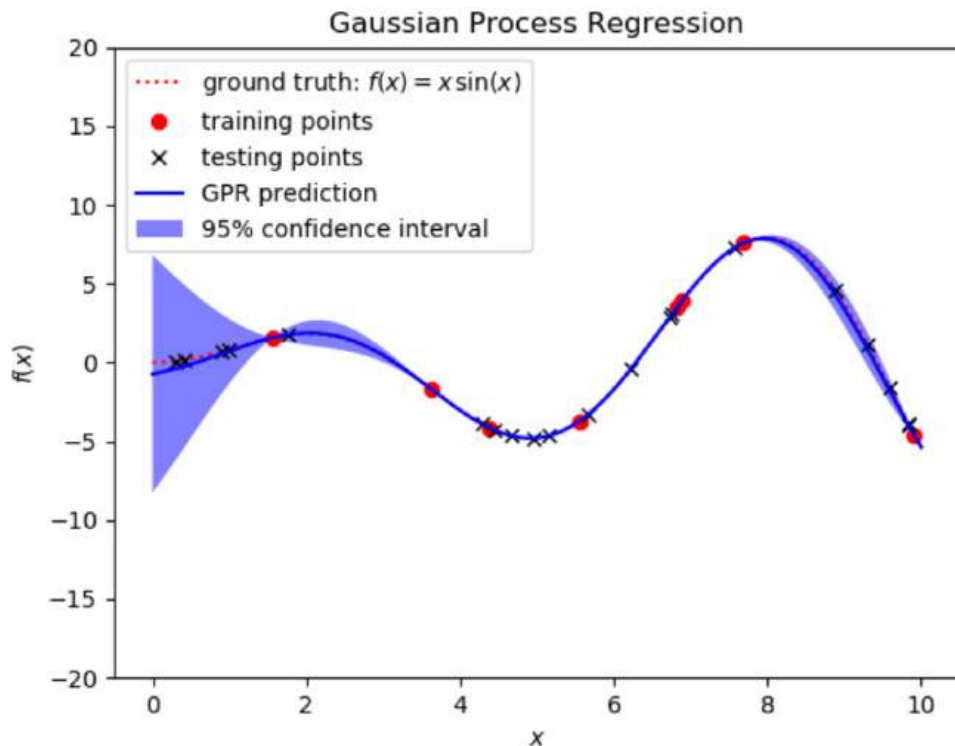
BO ?



Soon you'll be convinced that GP can be used for
solving Constrained Optimization problems

Surrogate GP vs NN

GPs offer superior uncertainty and performance in low-data, low/medium-dimensional engineering design tasks, while NNs excel in high-dimensional, complex, or data-rich regimes but require more tuning and approximate uncertainty.



M. Bessa, Machine learning in mechanics: simple resources, examples & opportunities.

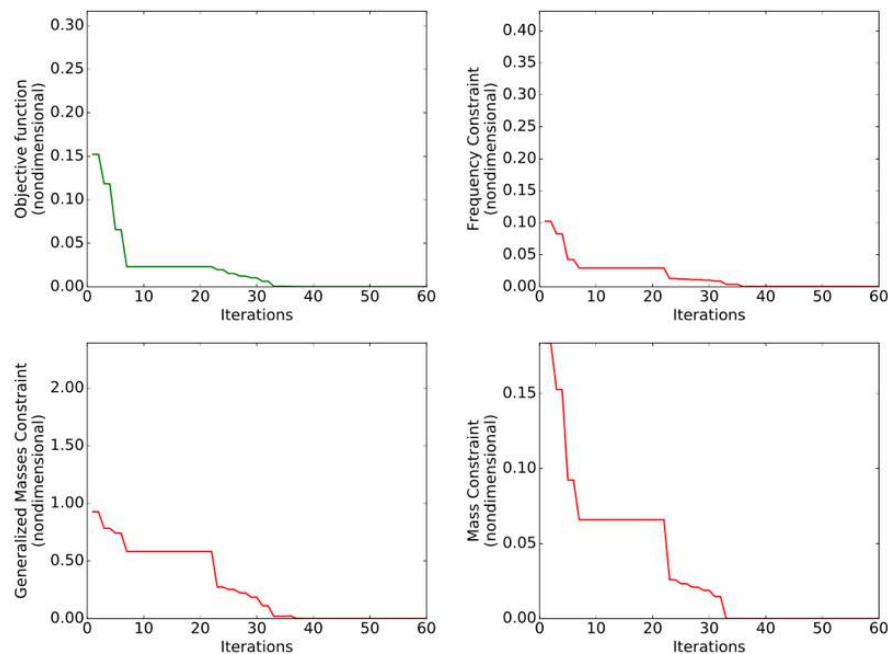
<https://imechanica.Org/node/23957>

By Mixing ...



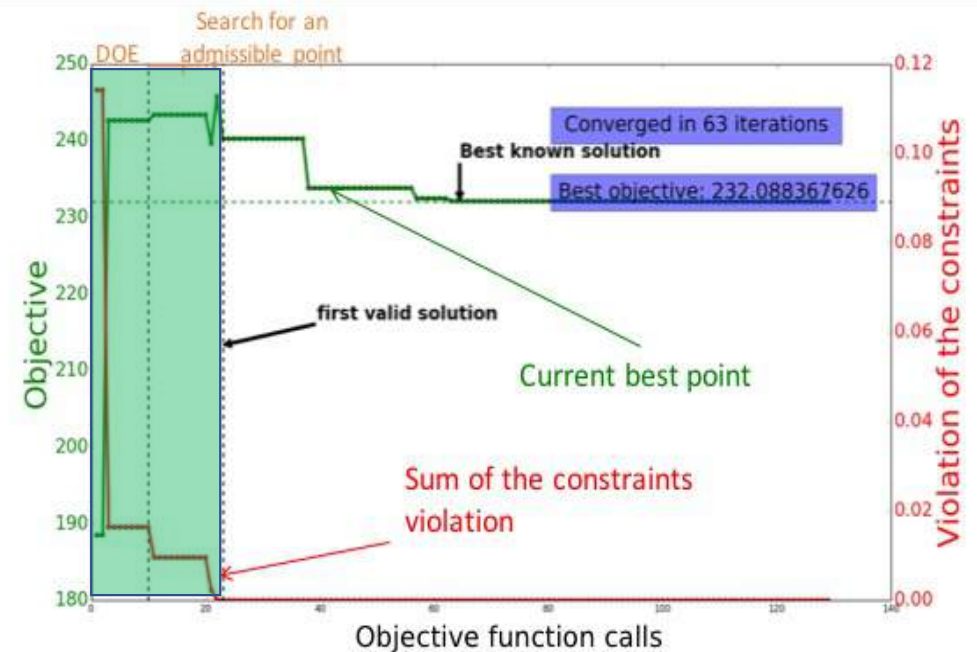
New paradigm for Bayesian Optimization (BO)

Gradient based Optimality, Feasibility



Stopping criteria: tolfun, tolx, maxiter

BO Exploration, Exploitation



Stopping criteria: Max Budget (Function calls)

Au programme



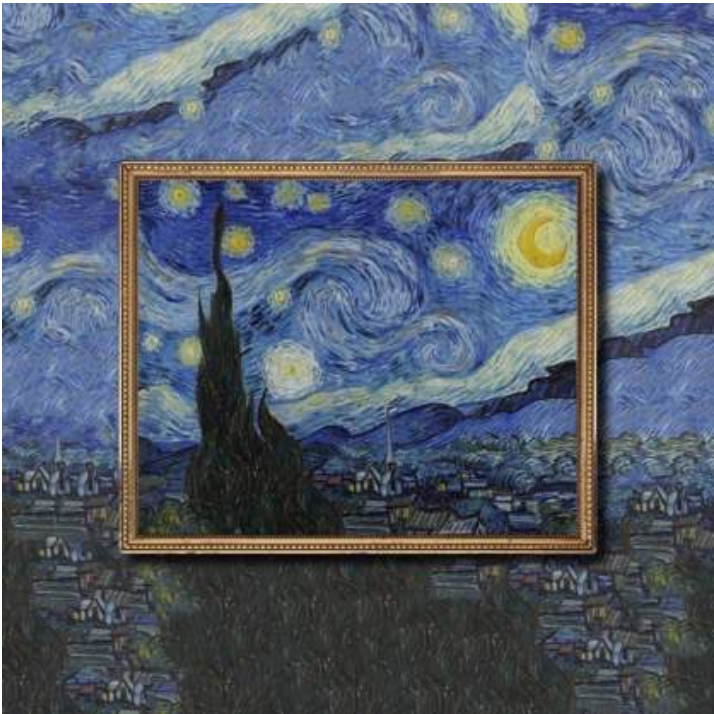
- Part1 : GP aka Kriging
- Part2: About SMT
- Part3: SEGOMOE and more

Part1 : GP aka Kriging

Mathematical foundations

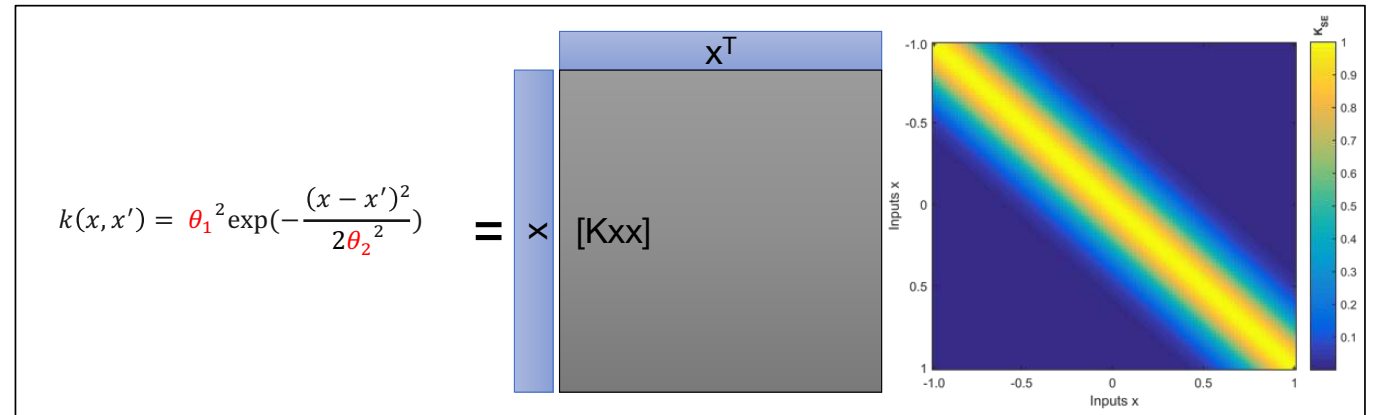
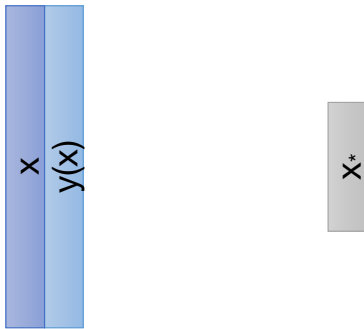
ML vs Engineering

Kriging (Pioneer)	Gaussian Processes (link with AI)
Developed by Daniel Krige – 1951; formalized by Georges Mathéron in the 60's (Mines Paris)	Neural network with infinite neurons tend to Gaussian Process 1994
Krige, D. G., 1951, A statistical approach to some basic mine valuation problems on the Witwatersrand: J. Chem. Metal. Min. Soc. South Africa, v. 52, p. 119-139. Matheron, G., 1963b, Principles of geostatistics: Economic Geol., v. 58, p. 1246-1266.	Neal, R. Priors for infinite networks. Tech. rep., University of Toronto, 1994. Williams, C. K. I., and Rasmussen, C. E. Gaussian processes for regression. <i>Advances in Neural Information Processing Systems 8</i> (1996), 514-520.



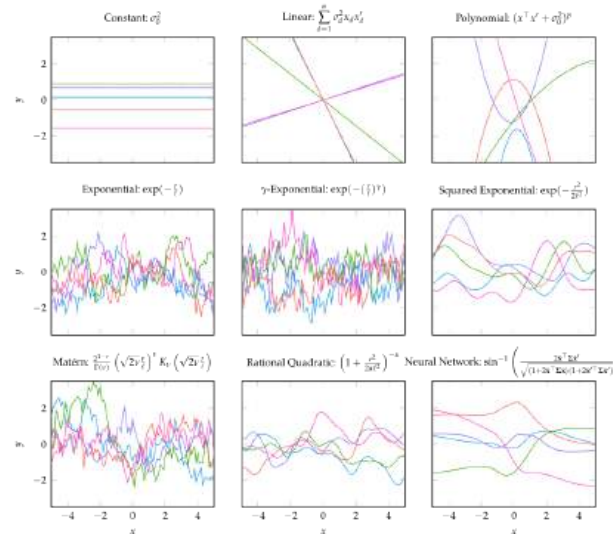
<http://extrapolated-art.com>

Matrix view of Gaussian Process



1/ Get your inputs/outputs data

2/ You want to predict at x^*



3/ Choose a Kernel
(it is not easy!, but you can cook it, for extrapolation, singularities, frequency content etc...)

<https://www.cs.toronto.edu/~duvenaud/cookbook>

Matrix view of Gaussian Process

1/ Get your inputs/outputs data

2/ You want to predict at $\mathbf{x}_s$

$\mathbf{x}$
 $\mathbf{y}(\mathbf{x})$

$\mathbf{x}_s$

$$k(\mathbf{x}, \mathbf{x}') = \theta_1^2 \exp\left(-\frac{(\mathbf{x} - \mathbf{x}')^2}{2\theta_2^2}\right)$$

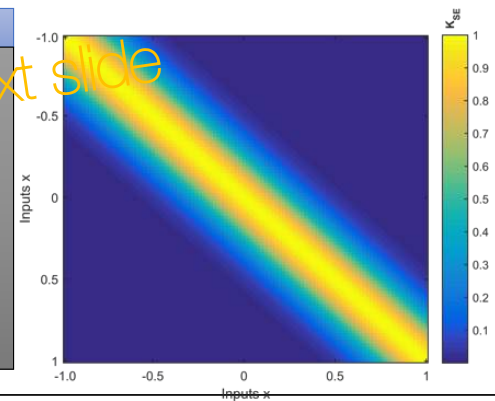
$= \times [\mathbf{K}_{xx}]$

3/ Construct $\mathbf{K}_{xx}$

4/ Tune your Hyperparameters

(How?)

Wait for next slide



$$\mathbf{m}(\mathbf{y}_*) = [\mathbf{K}_{\mathbf{x}_s \times \mathbf{x}_s}] [\mathbf{K}_{xx}]^{-1} \mathbf{y}(\mathbf{x})$$

`posterior_mean = covXXs @ np.linalg.inv(covXX_noisy) @ y`

4/ compute mean

$$\text{cov}(\mathbf{y}_*) = [\mathbf{K}_{\mathbf{x}_s \times \mathbf{x}_s}] - [\mathbf{K}_{\mathbf{x} \times \mathbf{x}_s}] [\mathbf{K}_{xx}]^{-1} [\mathbf{K}_{\mathbf{x} \times \mathbf{x}_s}]$$

Compute Interpolation
uncertainty (analytical)

`posterior_cov = covXsXs - covXXs @ np.linalg.inv(covXX_noisy) @`

Maximizing Marginal Likelihood (ML)

→ A third hyperparameter θ_3 or σ_n^2 is often used (noise) $GP(0, K + \sigma_n^2 I)$

$$ML = \log(p(y|X, \theta)) = -\frac{1}{2} y^T K^{-1} y - \frac{1}{2} \log|K| - \frac{n}{2} \log(2\pi)$$

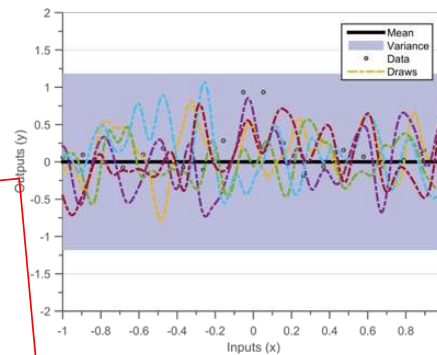
- It is a combination of **data-fit term**, a **complexity penalty** term and a **normalization term**

$$k(x, x') = \theta_1^2 \exp\left(-\frac{(x - x')^2}{2\theta_2^2}\right)$$

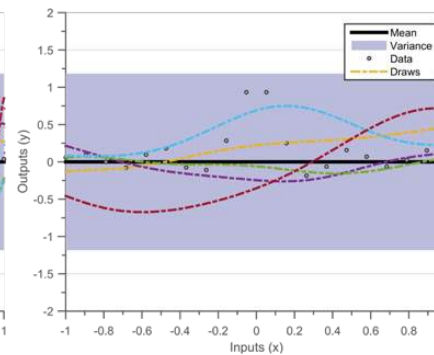
Only two hyperparameters:

- The lengthscale θ_2 or ℓ determines the length of the 'wiggles' in your function.
- The output variance θ_1^2 or σ^2 determines the average distance of your function away from its mean. It's just a scale factor.

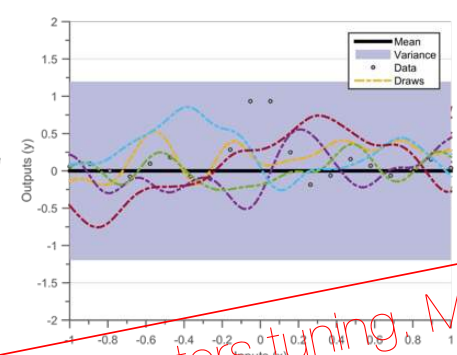
θ_1 and θ_2 often optimized using Cobyala



ML = -8.2



ML = -35.3



ML = 6.04

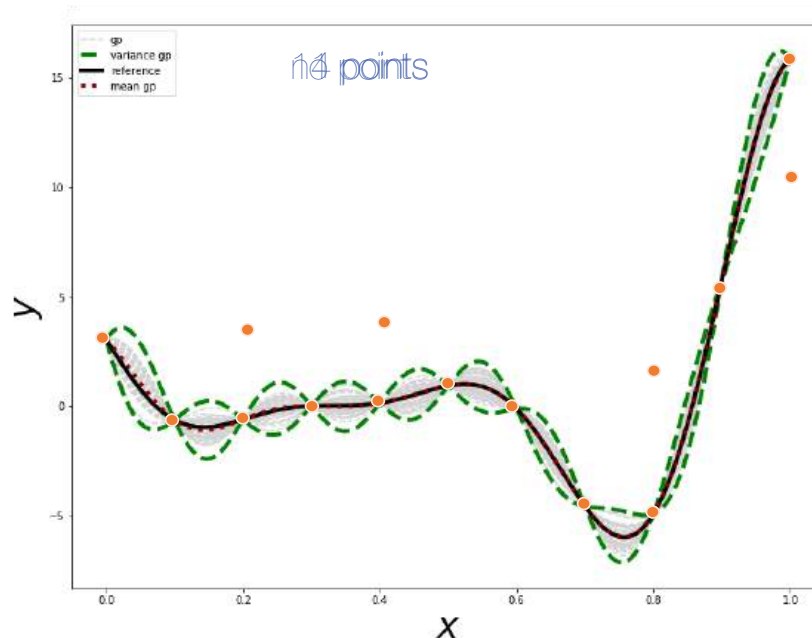
3/4 Hyperparameters tuning ML is maximised, θ^* is found

<https://docs.scipy.org/doc/scipy/reference/optimize.minimize-cobyala.html>

Gaussian process are interpolant (or not)

$$x \in \mathbb{R}^d$$

$$y(x) \in \mathbb{R}$$



Gaussian process characterized by:

- its mean (or trend)
 $\mu(x) \in \mathbb{R}$
 - its covariance kernel
 $k(x, x') \in \mathbb{R}$
- Estimation of hyperparameters $\theta_i (i=1, \dots, d)$ by MLE

Covariance matrix

+ information provided by data

Prediction $\hat{y}$ and variance $\hat{\sigma}^2$ estimations at a new point

- Hyperparameters tuning
 - Number of hyperparameters increases with the dimension d (number of design variables)
 - Curse of dimensionality
 - Size initial DOE wrt fixed budget of computations
- (99% of the time, the model is not interpolant)
- $$f(x) \Rightarrow Y(x) = \mathcal{N}(\hat{y}(x), s^2(x))$$

D. G. Krige, A statistical approach to some basic mine valuation problems on the Witwatersrand, 1951, Journal of the Southern African Institute of Mining and Metallurgy.

Part2: About SMT

X_0 in practice

Since 2017

<https://smt.readthedocs.io/en/latest>

<https://github.com/SMTorg/smt>



Required packages

SMT depends on the following modules: numpy, scipy, scikit-learn, pyDOE2 and Cython.

Installation

If you want to install the latest release

```
pip install smt
```

or else if you want to install from the current master branch

```
pip install git+https://github.com/SMTorg/smt.git@master
```

Usage

For examples demonstrating how to use SMT, you can take a look at the [tutorial notebook](#) or go to the 'smt/examples' folder.

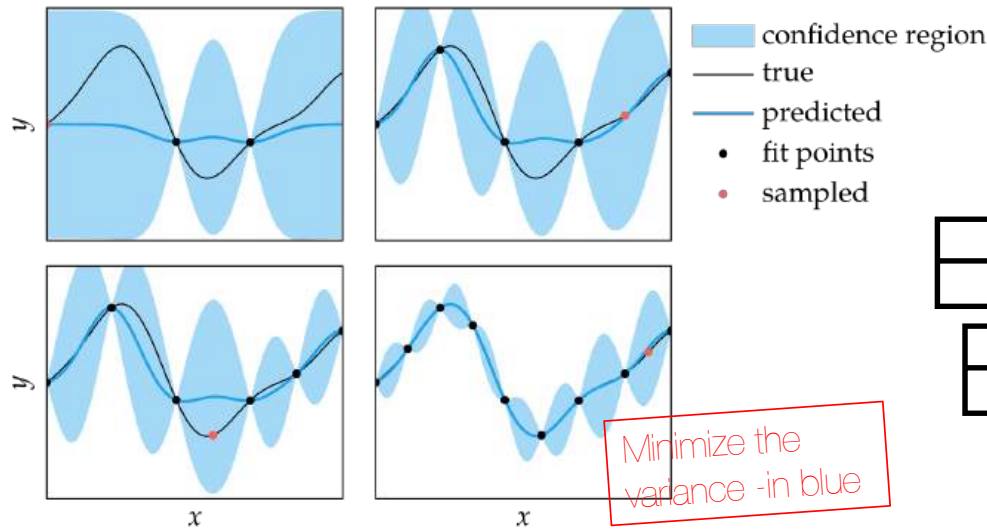
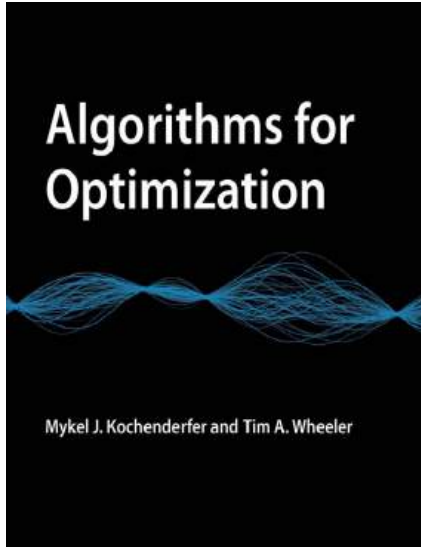
<https://github.com/SMTorg/smt/tree/master/tutorial#readme>



For this:
you need to understand one key concept

Given a surrogate model (GP) with both prediction and confidence interval, an optimization procedure must balance the search for the expected optimal point and decreasing uncertainty

*In other words, the algorithm must **balance exploitation with exploration to find the global optimum***



I want the most precise surrogate

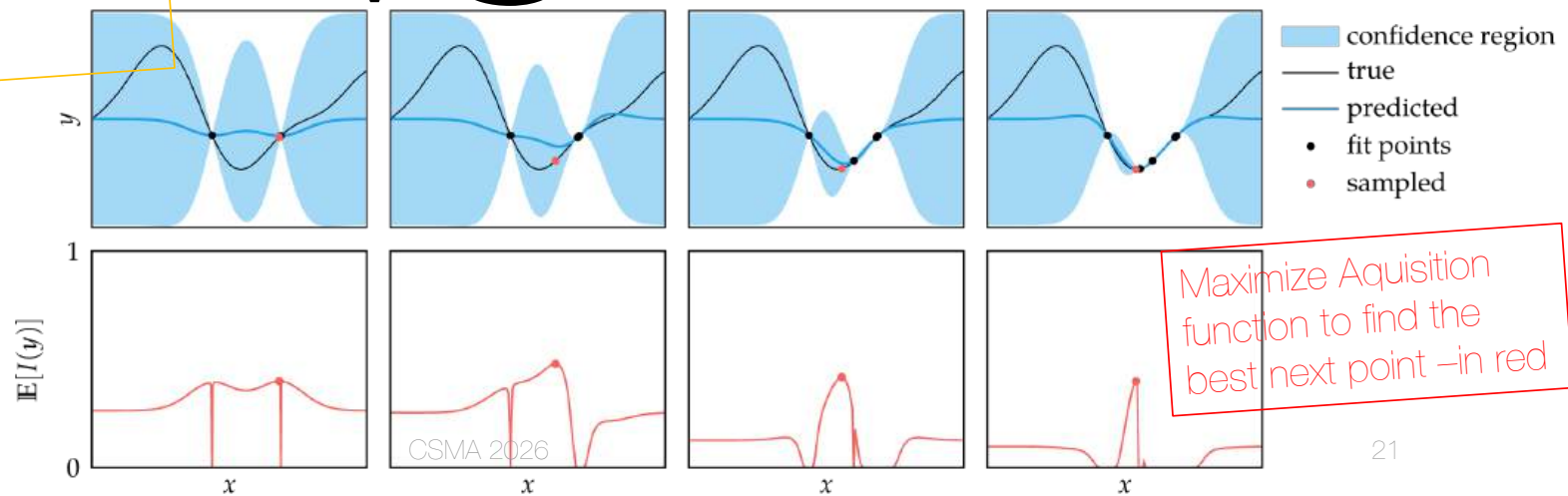
Error-Based Exploration

Minimize the variance - in blue

VS

I want a global optimum at fixed budget

Efficient Global Optimization



Maximize Acquisition function to find the best next point - in red

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Probability of Improvement Exploration

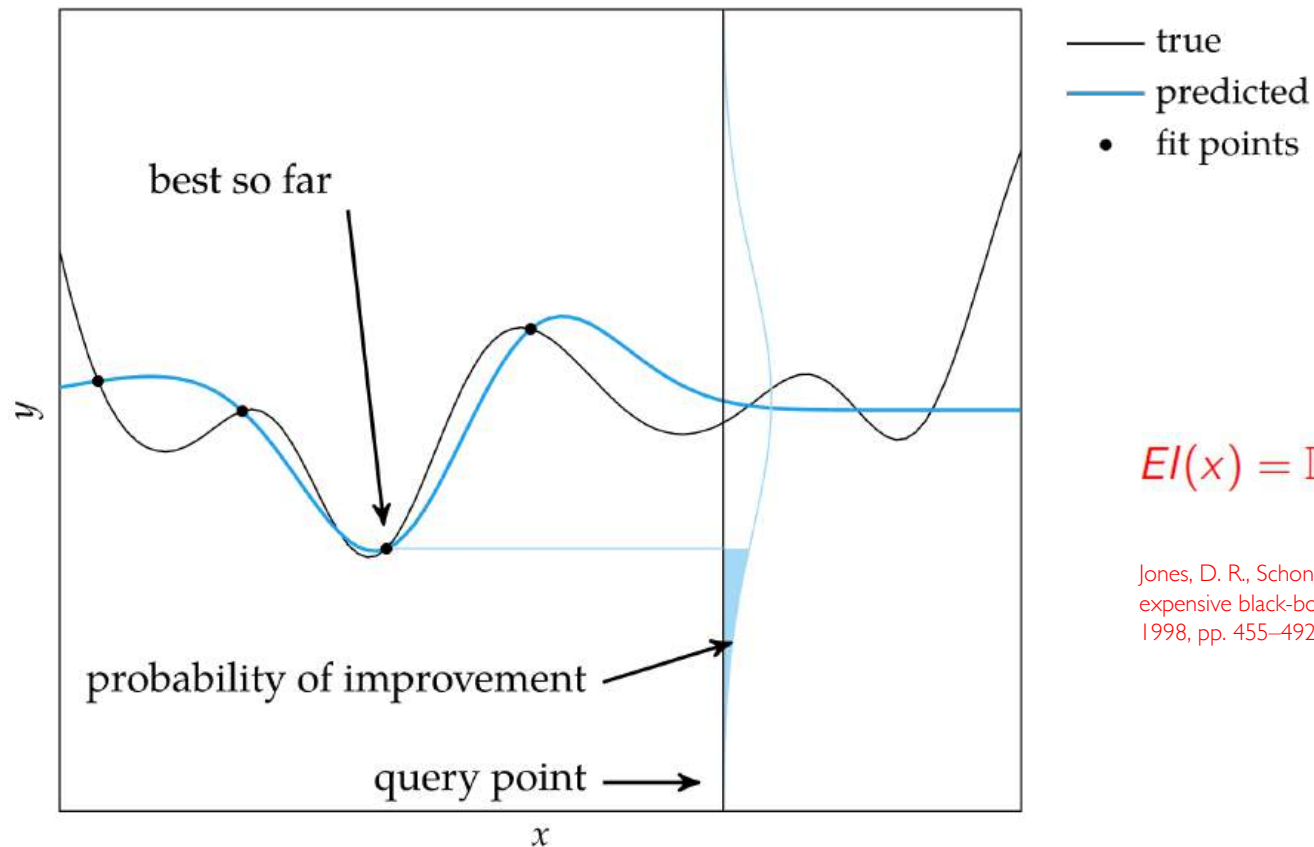
- Searches at the location with the highest probability of improvement
- Improvement is defined as

$$I(y) = \begin{cases} y_{\min} - y & \text{if } y < y_{\min} \\ 0 & \text{otherwise} \end{cases}$$

- The probability of improvement is defined as

$$\begin{aligned} P(y < y_{\min}) &= \int_{-\infty}^{y_{\min}} \mathcal{N}(y \mid \hat{\mu}, \hat{\sigma}) dy \\ &= \Phi\left(\frac{y_{\min} - \hat{\mu}}{\hat{\sigma}}\right) \end{aligned}$$

Probability of Improvement / Acquisition Function



$$EI(x) = \mathbb{E}[I(x)] = \mathbb{E}[\max\{0, y_{\min} - Y(x)\}]$$

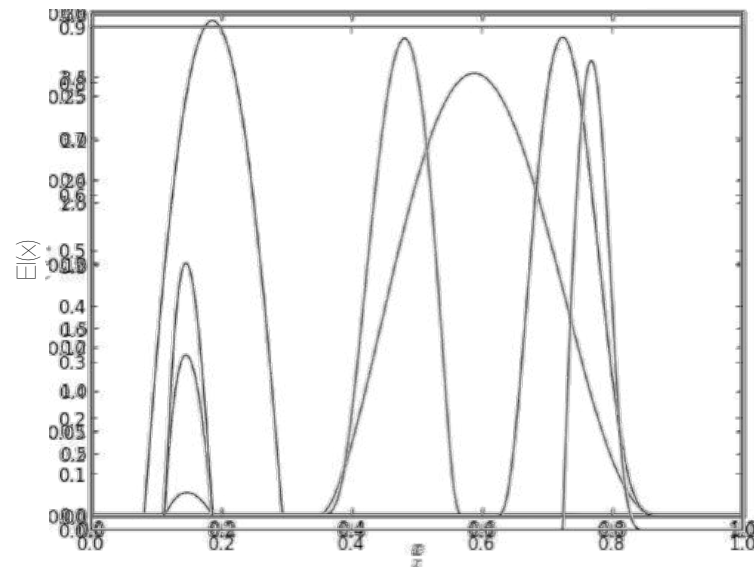
Jones, D. R., Schonlau, M., and Welch, W. J., "Efficient global optimization of expensive black-box functions," *Journal of Global optimization*, Vol. 13, No. 4, 1998, pp. 455–492.

Efficient Global Optimization

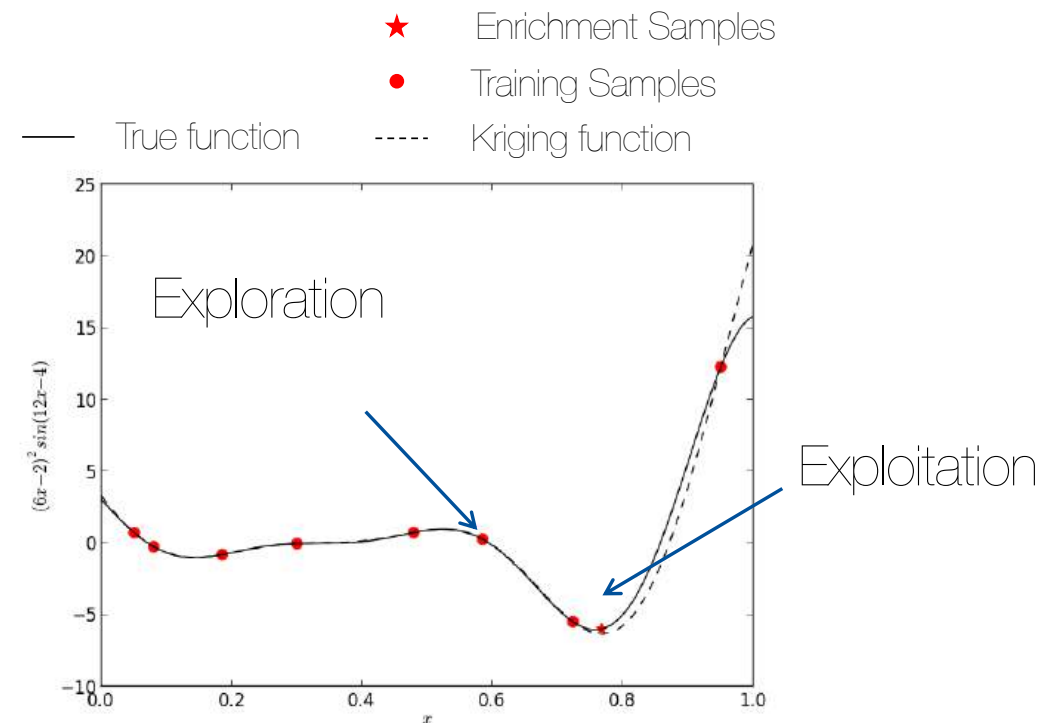
Next point * maximize EI

$$\begin{cases} \min (6x - 2)^2 \sin(12x - 4) \\ \text{s.t.} \\ 0 \leq x \leq 1 \end{cases}$$

$$EI(x) = \mathbb{E}[I(x)] = \mathbb{E}[\max\{0, y_{\min} - Y(x)\}]$$



- 4 points for the initial DOE
- 6 iterations



Mixed-type:

One-hot encoding:

Red: [1,0,0]

Green: [0,1,0]

Blue: [0,0,1]

Categorical, continuous, discrete, binary variables or Mixed

Categorical: Red, Green, Blue

Continuous: [0,1]

Discrete: 1,2,3,4,5

Binary: Yes, No

Tuning hyperparameters for deep neural network



$$= f(\underbrace{x_1, x_2}_{\substack{\text{continuous variables} \\ \text{learning rate } \in [1e^{-6}, 1e^{-1}] \quad \text{weight decay } \in [1e^{-6}, 1e^{-1}]}}, \underbrace{x_3, x_4}_{\substack{\text{categorical variables} \\ \text{optimiser type } \in \{SGD, Adam, \dots\} \quad \text{activation type } \in \{tanh, sigmoid, \dots\}}})$$

Multiple categorical - each categorical has multiple options

SOLUTION: A kernel-based approach that extends continuous exponential kernels to handle mixed-categorical variables (generalization of both the continuous relaxation and the Gower distance)

[A mixed-categorical correlation kernel for Gaussian process](#)

P. Saves, Y. Diouane, N. Bartoli, T. Lefebvre, J. Morlier
Neurocomputing 550, 126472

Mixed GP

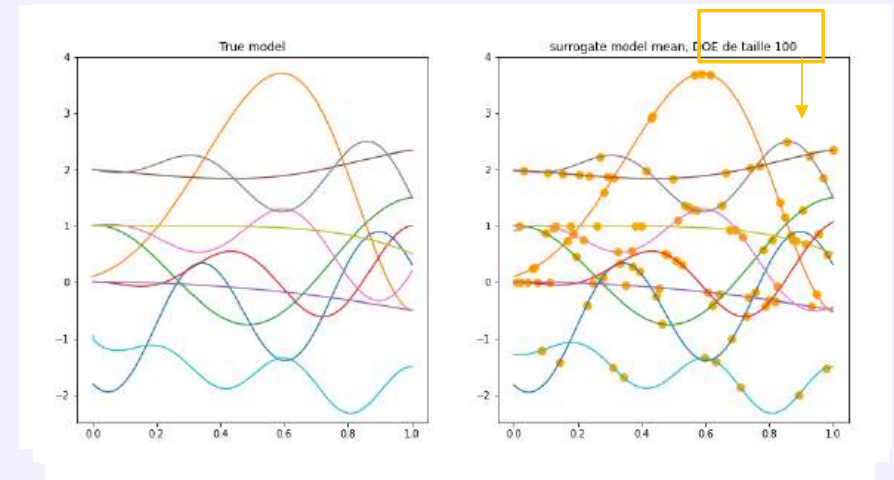
Validation problem $n = 2$

- Variable types: continuous and categorical with 10 levels, $n' = 11$

$$f(x, z) = \begin{cases} \cos(3.6\pi(x-2)) + x - 1 & \text{if } z = 1, \\ 2 \cos(1.1\pi \exp(x)) - \frac{x}{2} + 2 & \text{if } z = 2, \\ \cos(2\pi x) + \frac{1}{2}x & \text{if } z = 3, \\ x \left(\cos(3.4\pi(x-1)) - \frac{x-1}{2} \right) & \text{if } z = 4, \\ -\frac{x^2}{2} & \text{if } z = 5, \\ 2 \cos\left(\frac{\pi}{4} \exp(-x^4)\right)^2 - \frac{x}{2} + 1 & \text{if } z = 6, \\ x \cos(3.4\pi x) - \frac{x}{2} + 1 & \text{if } z = 7, \\ x \left(-\cos\left(7\frac{\pi}{2}x\right) - \frac{x}{2} \right) + 2 & \text{if } z = 8, \\ -\frac{x^5}{2} + 1 & \text{if } z = 9, \\ -\cos\left(5\frac{\pi}{2}x\right)^2 \sqrt{x} - \frac{\ln(x+0.5)}{2} - 1.3 & \text{if } z = 10. \end{cases}$$

Validation problem $n = 2$

- Variable types: continuous and categorical with 10 levels, $n' = 11$



Part3: SEGOMOE Engineering applications



botorch
Bayesian optimization in PyTorch (by pytorch)



Ax
Adaptive Experimentation Platform (by facebook)



https://scikit-learn.org/stable/modules/gaussian_process.html

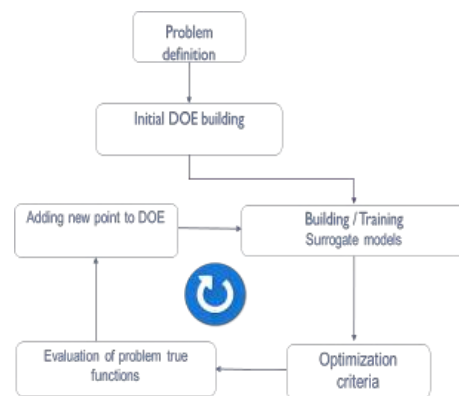
https://scikit-optimize.github.io/stable/auto_examples/bayesian-optimization.html

Engineering means constraints (a lot)

SEGOMOE (mono & multiobjective)

SEGO

Super Efficient Global
Optimization



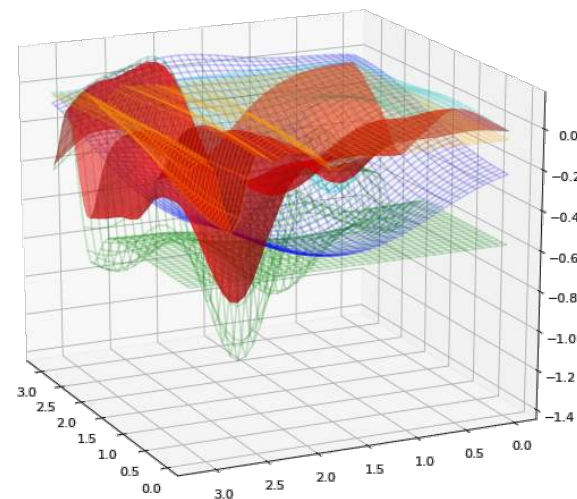
Global optimization with limited
number of function evaluations

Jones, D. R., Schonlau, M., and Welch, W. J., "Efficient global optimization of expensive black-box functions," *Journal of Global optimization*, Vol. 13, No. 4, 1998, pp. 455–492.

Sasena, M., Flexibility and efficiency enhancements for constrained global design optimization with Kriging approximations, Ph.D. thesis, University of Michigan, 2002

MOE

Mixture Of Experts



Combination of surrogate
models

Jordan, M. I., Jacobs, R. A., "Hierarchical mixtures of experts and the EM algorithm", *Neural Comput.* 6 (1994) 181–214.

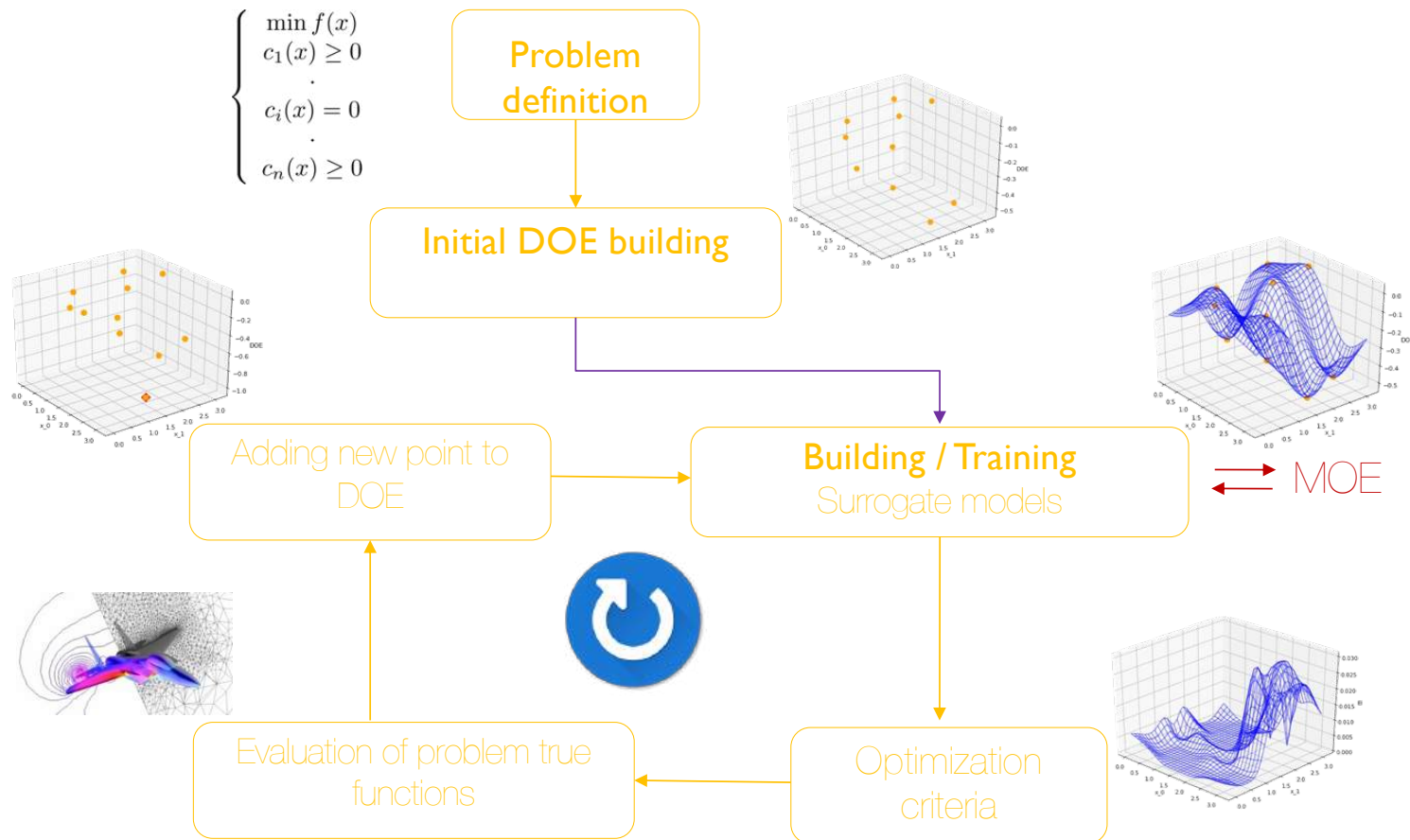
Bettebghor, D., Bartoli, N., Grihon, S., Morlier, J., and Samuelides, M., "Surrogate modeling approximation using a mixture of experts based on EM joint estimation," *Structural and Multidisciplinary Optimization*, Vol. 43, No. 2, 2011, pp. 243–259

Liem, R. P., Mader, C. A., and Martins, J. R. R. A., "Surrogate Models and Mixtures of Experts in Aerodynamic Performance Prediction for Mission Analysis," *Aerospace Science and Technology*, Vol. 43, 2015, pp. 126–151

SEGOMOE algorithm

N. Bartoli, T. Lefebvre, S. Dubreuil, R. Olivanti, N. Bons, J.R.R.A. Martins, M.-A. Bouhel, J. Morlier, "An adaptive optimization strategy based on mixture of experts for wing aerodynamic design optimization", 18th AIAA/ISSMO

Multidisciplinary Analysis Optimization Conference, AIAA-2017-4433, Denver, USA, June 2017



New formulation

Bartoli, N., Lefebvre, T., Dubreuil, S., Olivanti, R., Priem, R., Bons, N., ... & Morlier, J. (2019). Adaptive modeling strategy for constrained global optimization with application to aerodynamic wing design. *Aerospace Science and technology*, 90, 85-102.

Adapted from Super EGO

Costly initial problem

$$\begin{cases} \min f(x) \\ c_1(x) \geq 0 \\ \cdot \\ c_i(x) = 0 \\ \cdot \\ c_n(x) \geq 0 \end{cases}$$

Possibly Multimodal



Cheap enrichment problem

$$\begin{cases} \max_{x \in \mathbb{R}^d} EI(x)/WB2(x)/WB2s(x) \\ \text{s.t.} \\ \hat{c}_1(x) \geq 0 \\ \cdot \\ \hat{c}_i(x) = 0 \\ \cdot \\ \hat{c}_n(x) \geq 0 \end{cases}$$

n + 1
metamodels

Multimodal

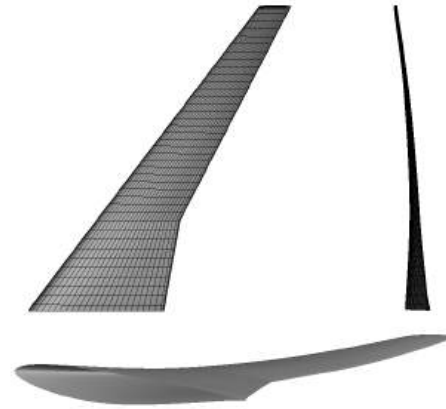
Global optimization method



Aerodynamic Shape Optimization

Simplified version of [ADODG Case 4](#) based on the CRM wing

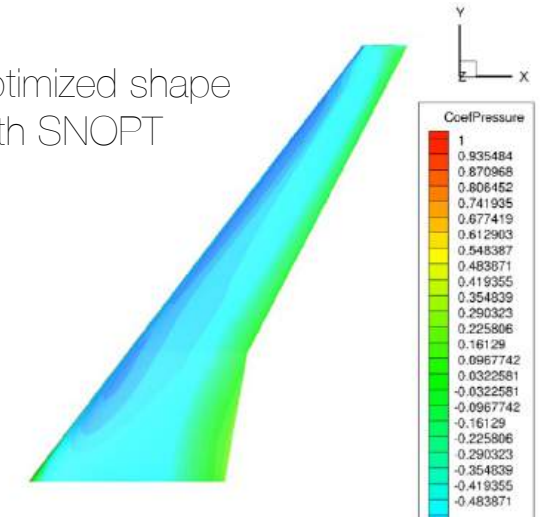
$$\begin{cases} \min & \text{Drag coefficient, } C_D \times 10^4 \\ \text{w. r. t.} & 8 \text{ twist variables } \in [-10, 10] \\ \text{s. t.} & C_L = 0.5 \end{cases}$$



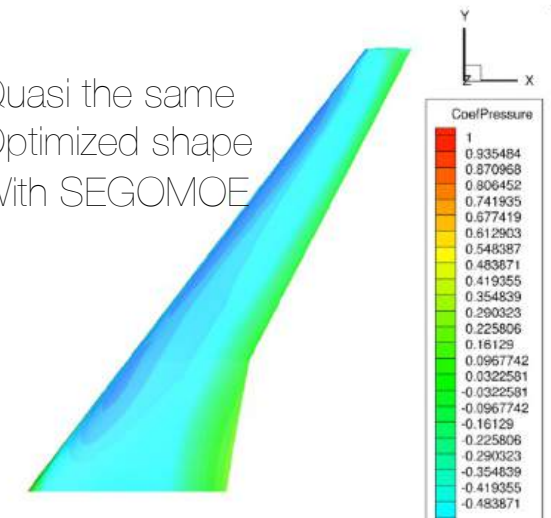
Optimizer	Best Objective	# of calls (direct and adjoint)
SNOPT	232.0884	87 (49+48)
NLPQL	232.0993	103 (52+51)

→ Fast and efficient: Convergence of SEGOMOE to the global optimum

Optimized shape
With SNOPT



Quasi the same
Optimized shape
With SEGOMOE

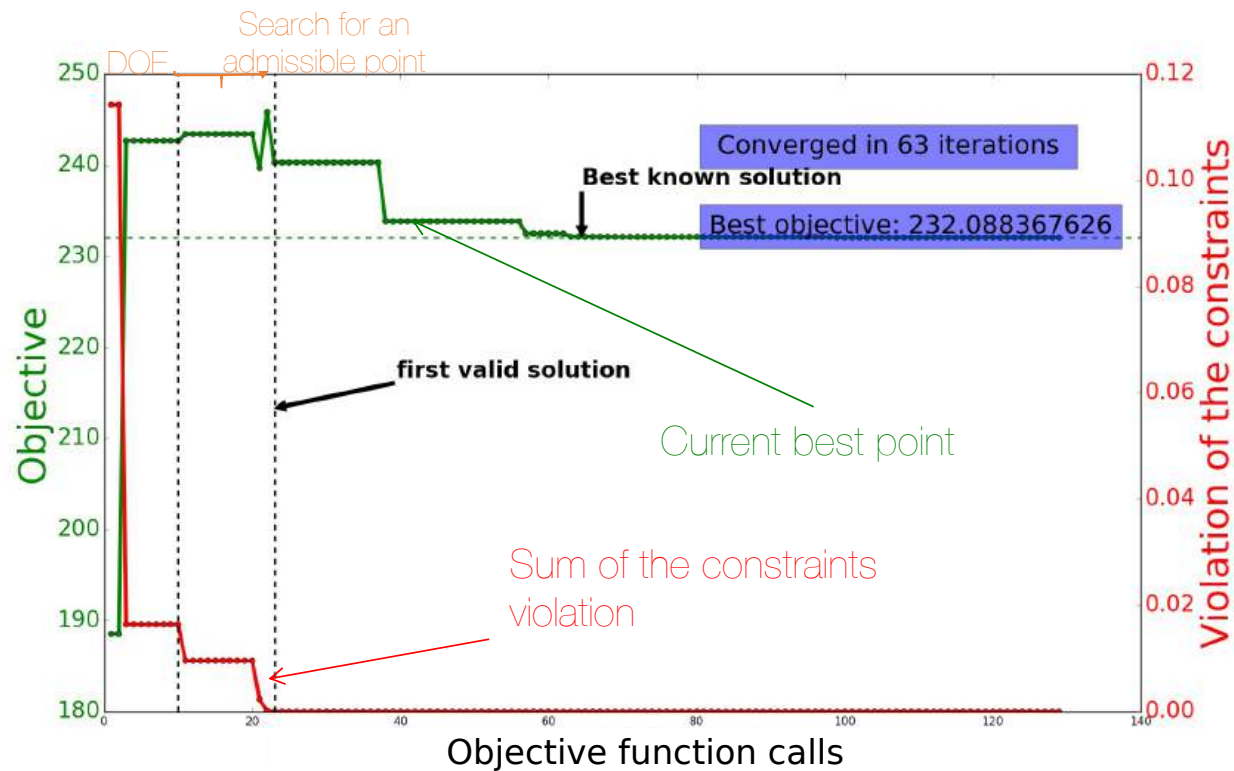


SEGOMOE results: objective function

Initial 9-points DOE

First steps to find an admissible point (constraints satisfied)

Then iterative process to reduce the objective function

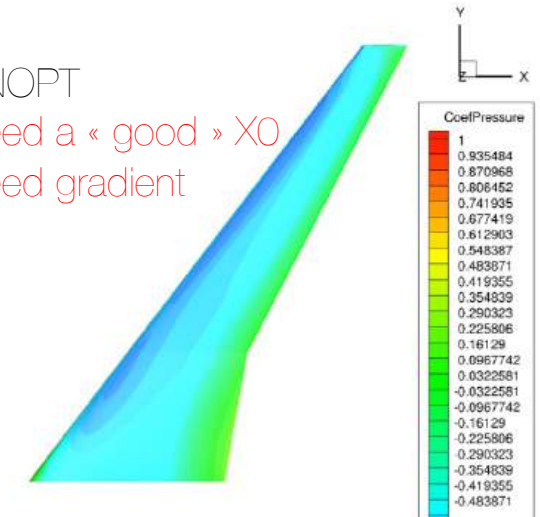


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SNOPT

Need a « good » XO

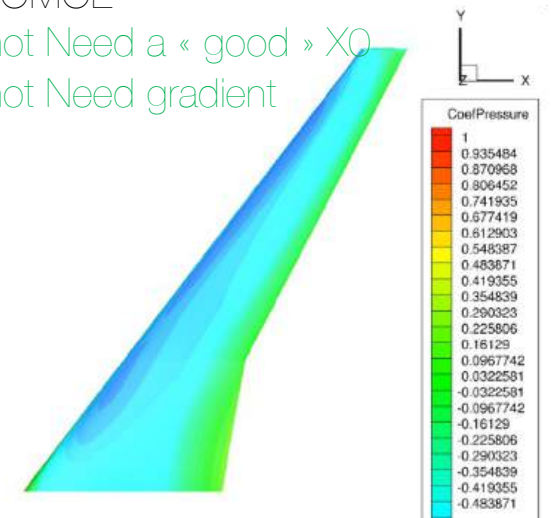
Need gradient



SEGOMOE

Do not Need a « good » XO

Do not Need gradient



36

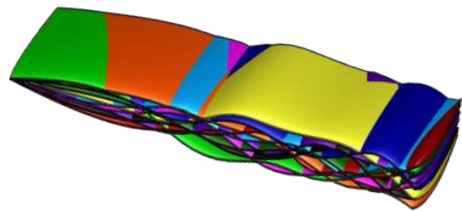
ADODG is really multimodal

Idea: 2 steps method

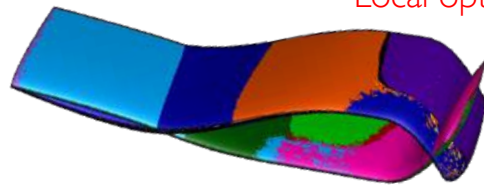
1. Start with SEGOMOE, stop with maxiter

$\mathbf{x}^\circ$ with high confidence near $\mathbf{x}^*$ (so we avoid to get stuck in local minima !)

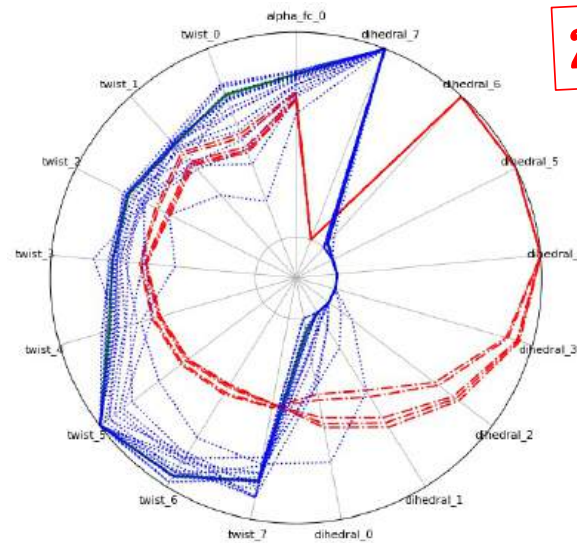
X_0 : Multi start with DOE of 15 (LHS)



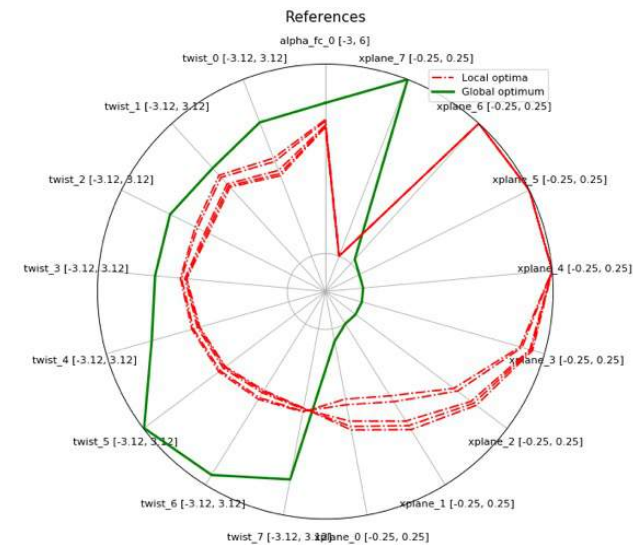
Local optima



Global optimum

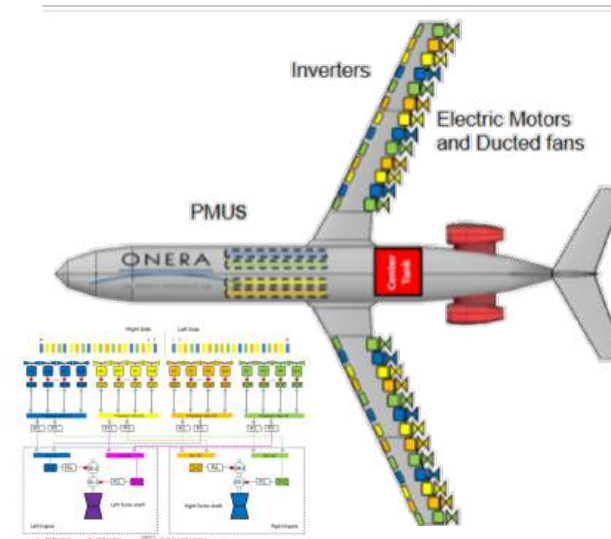


2. Use $\mathbf{x}^\circ$ as $\mathbf{x}_0$ in SNOPT to reach rapidly the Global Optimum $\mathbf{x}^*$



MDO

Optimization for new configurations



min objective function $f(x, y(x))$
with respect to design variables x (continuous, discrete, categorical, hierarchical)
subject to constraints $g(x, y(x))$
→ $y(x)$ are solution of a non linear system (MDA)
→ **Objective and constraint functions could be costly**

Saves, P., Bartoli, N., Diouane, Y., Lefebvre, T., Morlier, J., David, C., ... & Defoort, S. (2022). Bayesian optimization for mixed variables using an adaptive dimension reduction process: applications to aircraft design. In AIAA SCITECH 2022 Forum (p. 0082).

DRAGON

green aircraft concept

FAST OAD

Future Aircraft Sizing Tool - Overall Aircraft Design

<https://github.com/fast-aircraft-design/FAST-OAD>



	Function/variable	Nature	Quantity	Range
Minimize	Fuel mass	cont	1	
	Total objectives		1	
with respect to	Fan operating pressure ratio	cont	1	[1.05, 1.3]
	Wing aspect ratio	cont	1	[8, 12]
	Angle for swept wing	cont	1	[15, 40] (°)
	Wing taper ratio	cont	1	[0.2, 0.5]
	HT aspect ratio	cont	1	[3, 6]
	Angle for swept HT	cont	1	[20, 40] (°)
	HT taper ratio	cont	1	[0.3, 0.5]
	TOFL for sizing	cont	1	[1800., 2500.] (m)
	Top of climb vertical speed for sizing	cont	1	[300., 800.] (ft/min)
	Start of climb slope angle	cont	1	[0.075., 0.15.] (rad)
	Total continuous variables		10	
2 possibilities	Turboshaft layout	cat	2 levels	{1,2}
	Architecture_cat	cat	17 levels	{1,2,3, ..., 15,16,17}
	Number of cores	int	1	{2,4,6}
	Number of motors*	int	1	{8,12,16,20,...,40}
	*graph-structure dependence to the core value			
subject to	Wing span < 36 (m)	cont	1	
	TOFL < 2200 (m)	cont	1	
	Wing trailing edge occupied by fans < 14.4 (m)	cont	1	
	Climb duration < 1740 (s)	cont	1	
	Top of climb slope > 0.0108 (rad)	cont	1	
	Total constraints		5	

- 10 continuous design variables
- 2 categorical design variables
 - Electric propulsion Architecture: 17 choices
 - Turboshaft layout: 2 choices
- 5 inequality constraints (MC)
- Fuel mass to minimize

17x2 possibilities → 34 associated optimization problems to solve ☹️
 or use dedicated kernels in your GP models → 1 optimization problem to solve 😊

A mixed-categorical correlation kernel for Gaussian process

P. Saves, Y. Diouane, N. Bartoli, T. Lefebvre, J. Morlier
 Neurocomputing 550, 126472

IDEA !

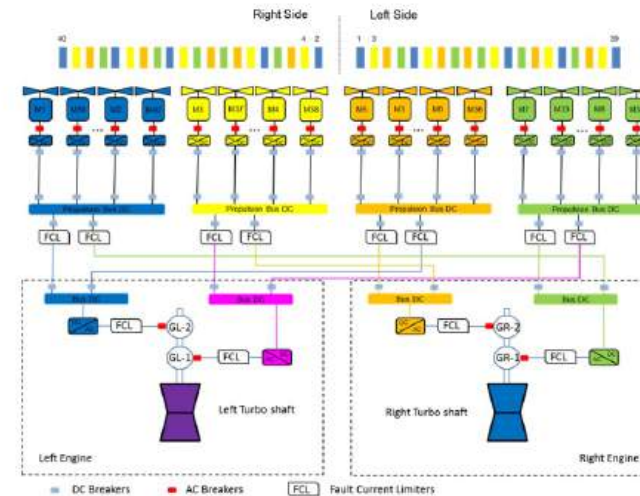
Architecture	cat	17 levels	{1,2,3,...,15,16,17}
Turboshaft layout	cat	2 levels	{1,2}
Total categorical variables		2	
Total relaxed variables		29	

Architecture number	number of generators	number of motors
1	2	8
2	2	12
3	2	16
4	2	20
5	2	24
6	2	28
7	2	32
8	2	36
9	2	40
10	4	8
11	4	16
12	4	24
13	4	32
14	4	40
15	6	12
16	6	24
17	6	36

Neutral

Meta

Decreed



layout	position	y ratio	tail	VT aspect ratio	VT taper ratio
1	under wing	0.25	without T-tail	1.8	0.3
2	behind	0.34	with T-tail	1.2	0.85

- 2 possibilities for dealing with categorical variables:

→ Categorical choice:

29 variables in relaxed dimension (10+17+2)

→ Hierarchical choice:

14 variables in relaxed dimension (10+2+2)

- Need dedicated kernels (with adapted distance)

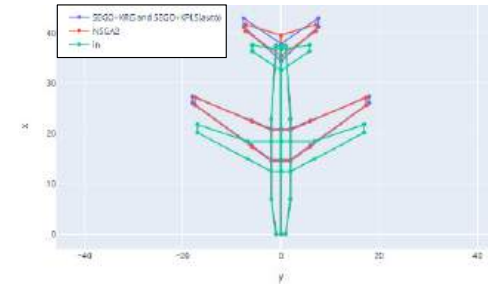
A mixed-categorical correlation kernel for Gaussian process

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Neurocomputing 550, 126472

Comparison of BO with different kernels & NSGAII

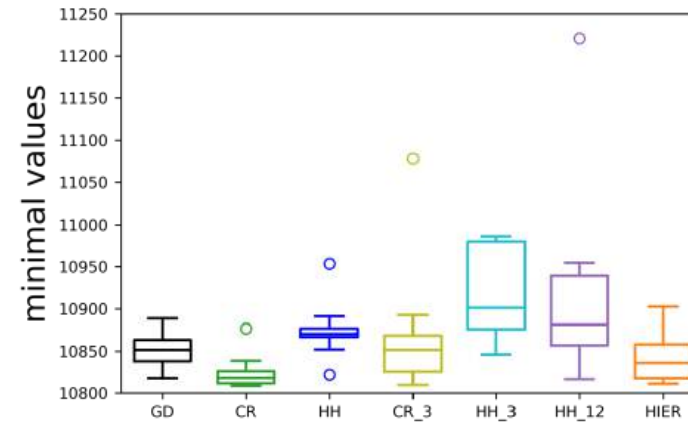
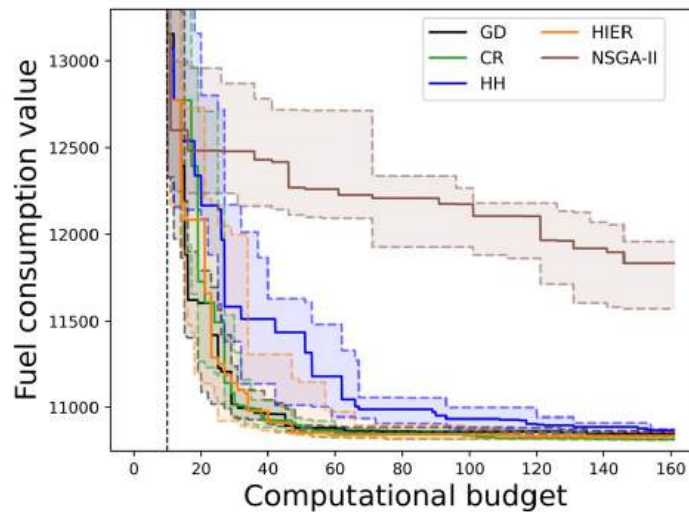
10 runs of (10 + 150) iterations



Convergence plots

Categorical or Hierarchical variants

Boxplots
after 160 evaluations



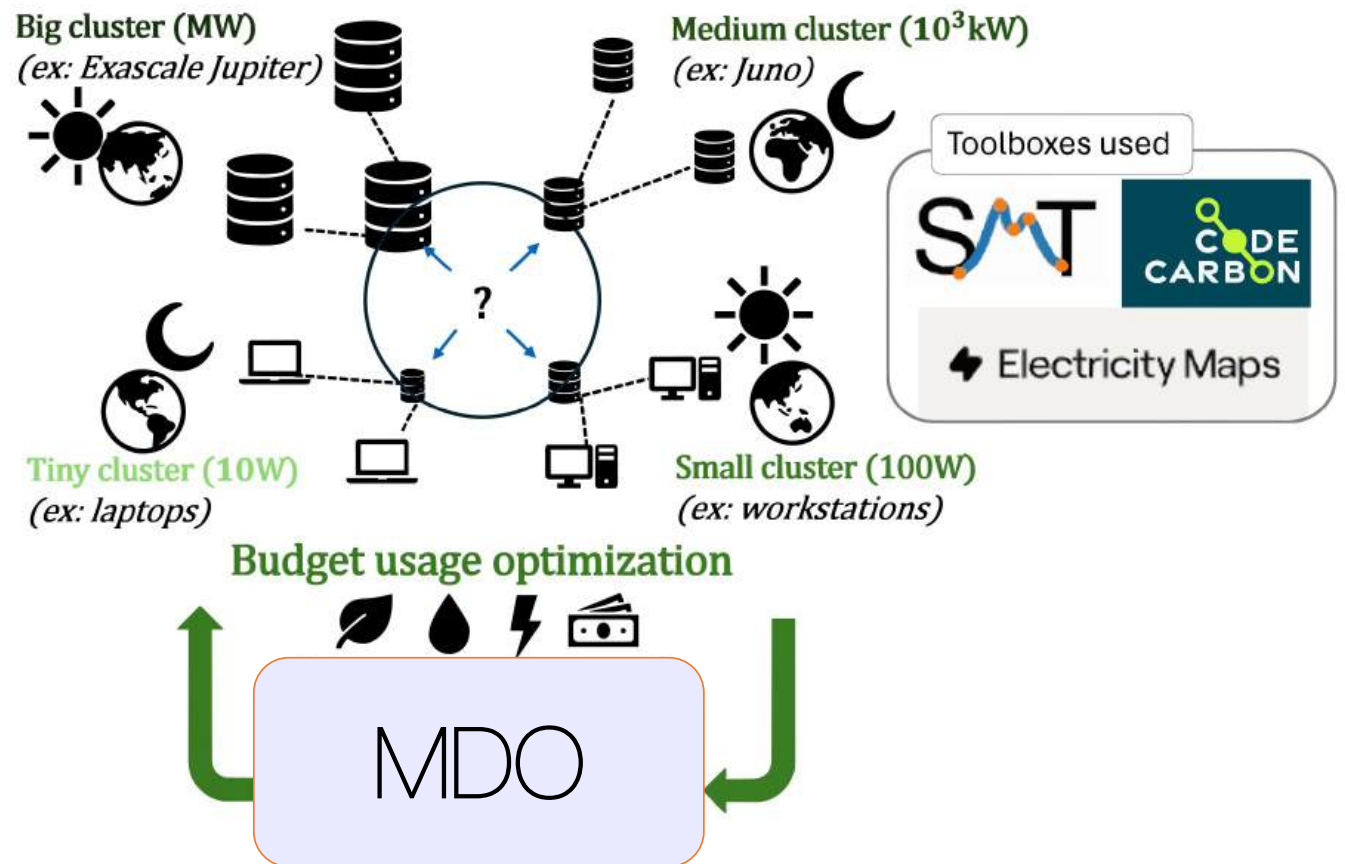
Gower distance GD
Continuous Relaxation CR
Homoscedastic hypersphere HH

→ Hierarchical choice: best trade off convergence & CPU time

Saves, P., Bartoli, N., Diouane, Y., Lefebvre, T., Morlier, J., David, C., ... & Defoort, S. (2022). Bayesian optimization for mixed variables using an adaptive dimension reduction process: applications to aircraft design. In AIAA SCITECH 2022 Forum (p. 0082).

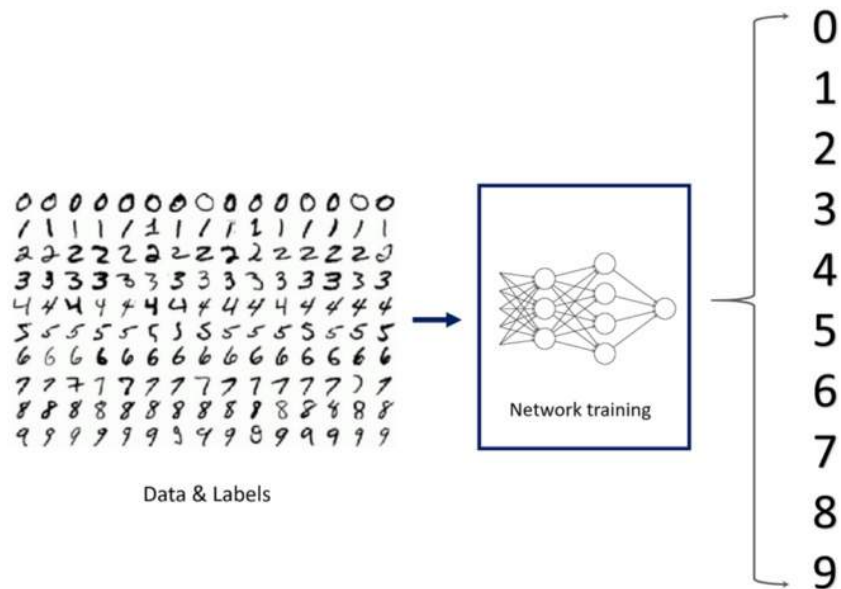
MOO

- **NAS** Thanks to our PhD Gaston Plat



Frugal AI (2 objectives)

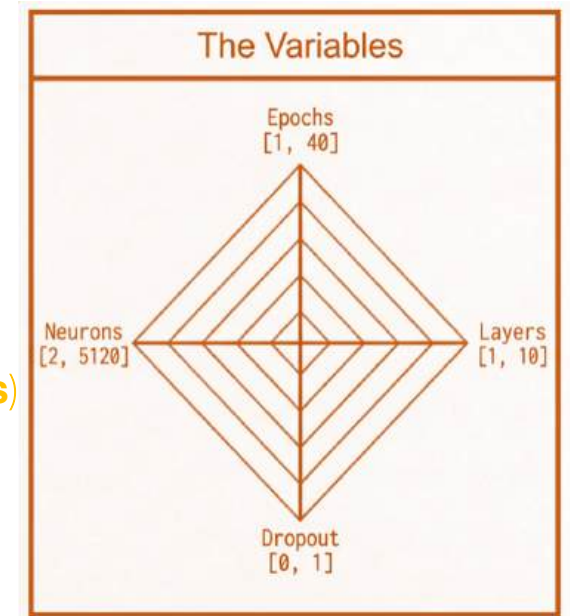
well established MNIST problem



Min (energy, error)

w.r.t. NN parameters

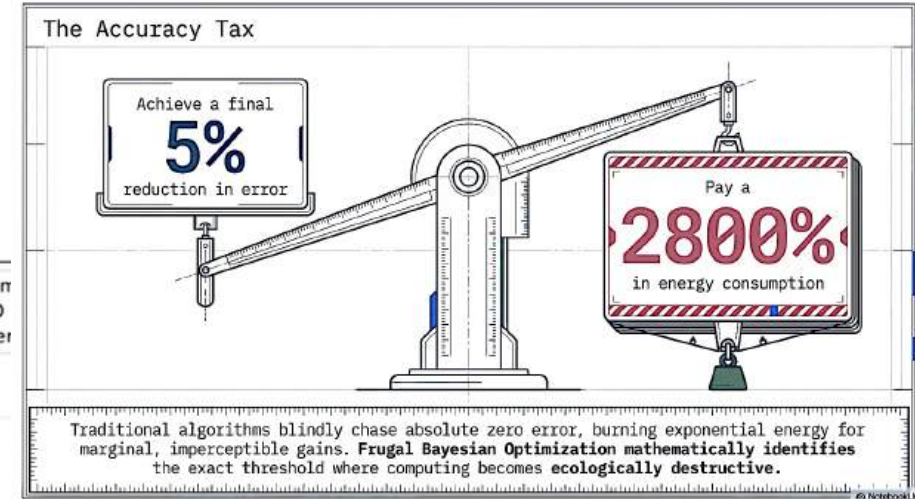
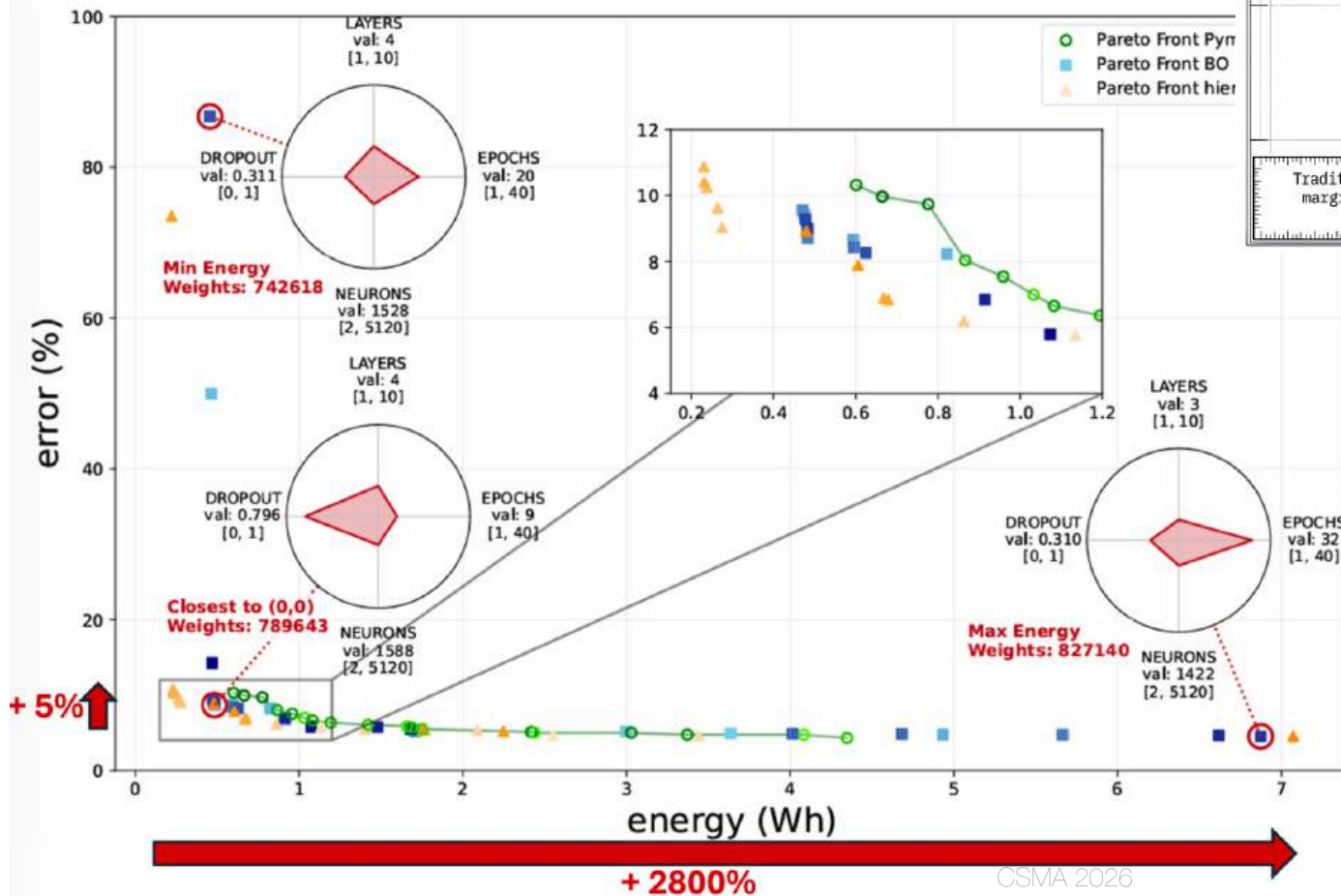
(epochs, layers, dropout, neurons)



2 Methods: SEGOMOE vs NSGA II

Comparison with SoA

200 BO vs 700 it for NSGA II



Conclusions

And more

Conclusions

« Learning » an industrial (&costly) simulation code is interesting to easily exchange data only (without having access to the code in a collaborative project)



SEGOMOE framework is **one efficient solution** for Constrained Bayesian Optimization (DV<100 through KPLS).

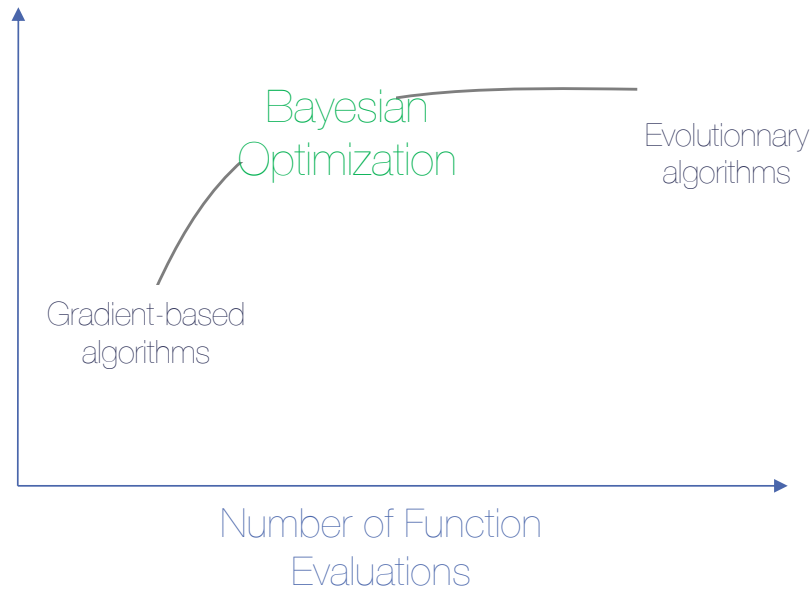
It relies on **SMT core capabilities** for Explainability/Kernel cooking, mixed variables, multifidelity, multiObjectives

Automate the choice of ML architecture : makes it easy to create accurate and efficient emulators for complex simulations. Under the hood, the package runs a complete machine learning pipeline to compare and optimise a wide range of models



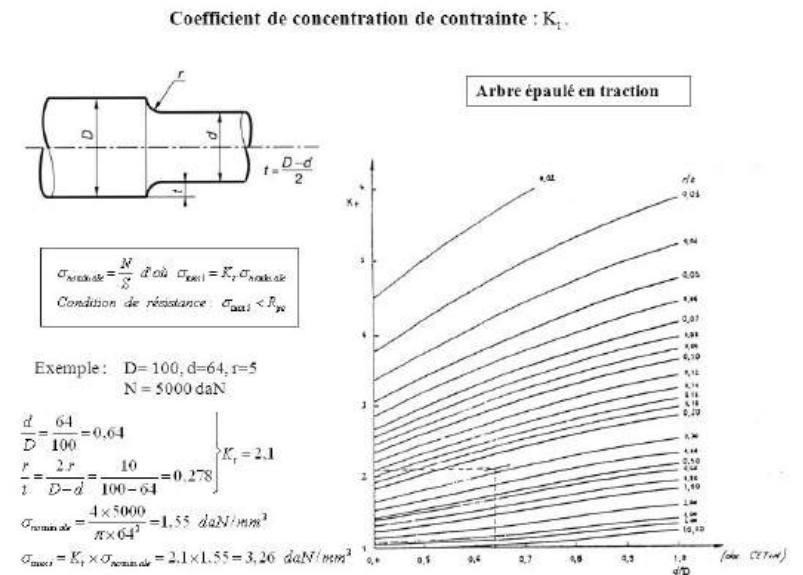
Conclusions

Global aspect



Want to test?

Surrogate is the new abacus



Constrained BO in opensource ?

Rust/efficient code based on SMT



README Apache-2.0 license



EGObox - Efficient Global Optimization toolbox

tests passing pytest passing lint passing JOSS 10.21105/joss.04737

Rust toolbox for Efficient Global Optimization inspired by [the EGO implementation](#) in the [SMT](#) Python library.

The `egobox` package is twofold:

1. for end-users: [a Python module](#), the Python binding of the optimizer named `Egor` and the surrogate model `Gpx`, mixture of Gaussian processes, written in Rust.
2. for developers: [a set of Rust libraries](#) useful to implement bayesian optimization (EGO-like) algorithms,

Opensource : <http://github.com/relf/egobox>

JOSS : <https://joss.theoj.org/papers/10.21105/joss.04737>

Installation : `pip install egobox`

`[+ pip install openmdao_extensions]`

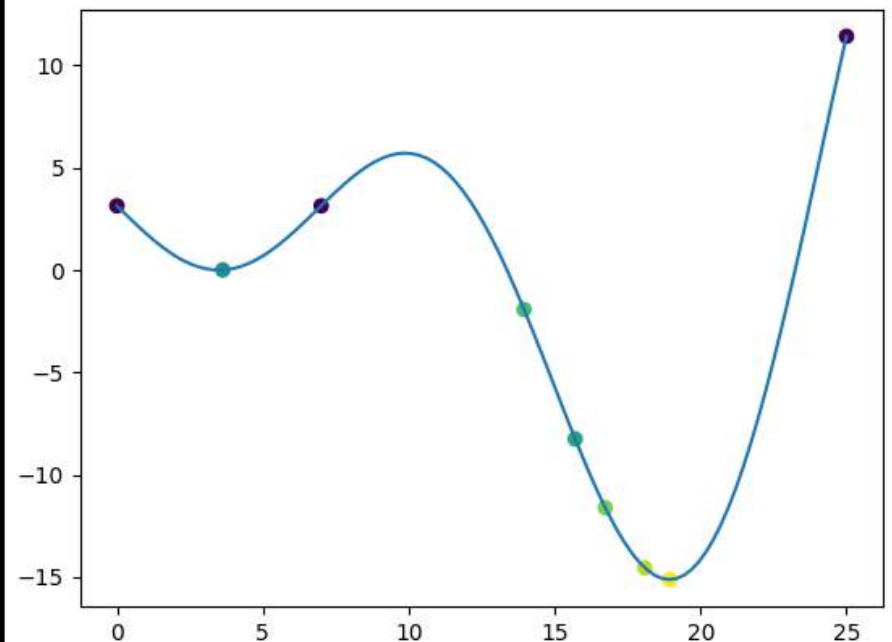
EGObox – Egor optimizer example

```
import numpy as np
import egobox as egx

# Objective function (npts, dimx) -> (npts, dimy)
def f_obj(x: np.ndarray) -> np.ndarray:
    return (x - 3.5) * np.sin((x - 3.5) / (np.pi))

# Minimize f_opt in [0, 25]
doe = np.array([[0.0], [7.0], [25.0]])
egor = egx.Egor([[0.0, 25.0]], # design space (xdim, 2)
                doe=doe, # initial doe (x only or x,y)
                infill_strategy=egx.InfillStrategy.El)
res = egor.minimize(f_obj, max_iters=6)

print(f"Optimization f={res.y_opt} at {res.x_opt}")
# Optimization f=[-15.12510323] at [18.93525454]
```



https://github.com/relf/egobox/blob/master/doc/Egor_Tutorial.ipynb



COLLABORATIVE OPTIMIZATION

- ONERA: P. Saves and R. Lafage and N. Bartoli , T. Lefevbre, R. Charayron...
- ISAE-SUPAERO: R. Rufato Careira, P. Saves, R. Charayron, G. Plat
- U. Capitole/IRIT: P. Saves
- ITB: P. Pallar and students
- Polytechnique Montréal: Y. Diouane and students
- DLR: J. H. Bussemaker
- UCSD: J. T. Hwang and students
- UoM: J. R. R. A. Martins and students
- SMT 1.0 original coding due to Mohamed Amine Bouhlel (INTEL)
- Many more SUPAERO's students !!!!!